Homework Set C.1

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Problem 1

$$\hat{i} \times \hat{j} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \hat{i}_x & \hat{i}_x & \hat{i}_x \\ \hat{j}_x & \hat{j}_x & \hat{j}_x \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \hat{i} \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = (0)\hat{i} - (0)\hat{j} + (1)\hat{k} = \boxed{\hat{k} = (0, 0, 1)}$$

Problem 2

$$\begin{array}{lll} \hat{i} \times \hat{i} = 0 & \hat{i} \times \hat{j} = \hat{k} & \hat{i} \times \hat{k} = -\hat{j} \\ \hat{j} \times \hat{i} = -\hat{k} & \hat{j} \times \hat{j} = 0 & \hat{j} \times \hat{k} = \hat{i} \\ \hat{k} \times \hat{i} = \hat{j} & \hat{k} \times \hat{j} = -\hat{i} & \hat{k} \times \hat{k} = 0 \end{array}$$

Problem 3

Cross product is anti-commutative: $a \times b = -b \times a$.

Proof. First, notice that the determinant of a matrix formed by swapping the rows of a 2×2 matrix is the negative determinant of the original matrix:

$$\begin{vmatrix} p & q \\ r & s \end{vmatrix} = ps - qr = -(qr - ps) = -(rq - sp) = - \begin{vmatrix} r & s \\ p & q \end{vmatrix}$$

Let a, b be vectors in $\mathbb{R}^3$ with the components $a = (a_x, a_y, a_z)$ and $b = (b_x, b_y, b_z)$.

$$a \times b = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \hat{i} \begin{vmatrix} a_y & a_z \\ b_y & b_z \end{vmatrix} - \hat{j} \begin{vmatrix} a_x & a_z \\ b_x & b_z \end{vmatrix} + \hat{k} \begin{vmatrix} a_x & a_y \\ b_x & b_y \end{vmatrix}$$

$$-b \times a = - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_x & b_y & b_z \\ a_x & a_y & a_z \end{vmatrix} = - \left(\hat{i} \begin{vmatrix} b_y & b_z \\ a_y & a_z \end{vmatrix} - \hat{j} \begin{vmatrix} b_x & b_z \\ a_x & a_z \end{vmatrix} + \hat{k} \begin{vmatrix} b_x & b_y \\ a_x & a_y \end{vmatrix} \right)$$

$$= \hat{i} \left(- \begin{vmatrix} b_y & b_z \\ a_y & a_z \end{vmatrix} \right) - \hat{j} \left(- \begin{vmatrix} b_x & b_z \\ a_x & a_z \end{vmatrix} \right) + \hat{k} \left(- \begin{vmatrix} b_x & b_y \\ a_x & a_y \end{vmatrix} \right)$$

Using the observation earlier, to find the negative determinant, we can swap the rows of the matrix:

$$= \hat{i} \begin{vmatrix} a_y & a_z \\ b_y & b_z \end{vmatrix} - \hat{j} \begin{vmatrix} a_x & a_z \\ b_x & b_z \end{vmatrix} + \hat{k} \begin{vmatrix} a_x & a_y \\ b_x & b_y \end{vmatrix} = a \times b$$

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